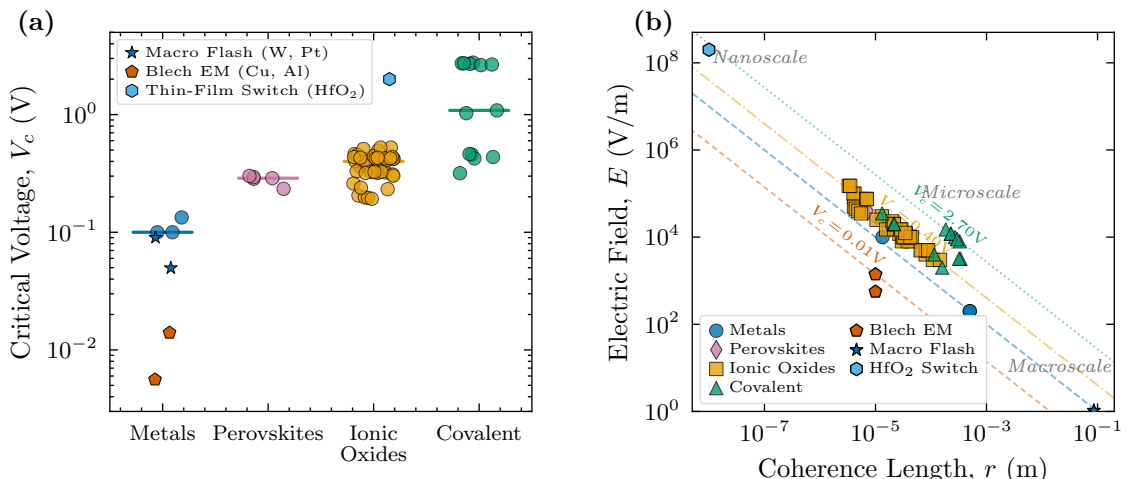
(a)
Macro Flash (W, Pt)
Blech EM (Cu, Al)
Thin-Film Switch (HfO2)
Critical Voltage, Vc (V)
10^0
10^-1
10^-2
Metals
Perovskites
Ionic Oxides
Covalent
(b)
Nanoscale
Microscale
Macroscale
Vc = 2.70V
Vc = 0.41V
Vc = 0.01V
Metals
Perovskites
Ionic Oxides
Covalent
Blech EM
Macro Flash
HfO2 Switch
Electric Field, E (V/m)
10^8
10^6
10^4
10^2
10^0
10^-7
10^-5
10^-3
10^-1
Coherence Length, r (m)